SYLLABUS : ELECTROMAGNETIC FORCE

1. A charged particle moves through a magnetic field perpendicular to its direction. Then :

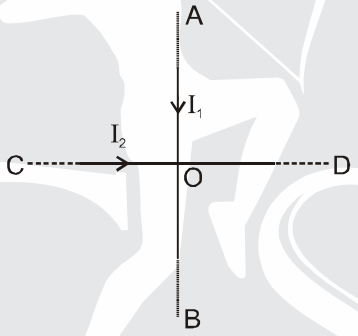
(A) the momentum changes but the kinetic energy is constant

(B) both momentum and kinetic energy of the particle are not constant

(C) both, momentum and kinetic energy of the particle are constant

(D) kinetic energy changes but the momentum is constant

2. Two identical conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries an electric current I_1 and COD carries a current I_2. The magnetic field on a point lying at a distance d from O, in a direction perpendicular to the plane of the wires AOB and COD, will be given by :



(A) $\frac{\mu_0}{2\pi}\left(\frac{I_1+I_2}{d}\right)^{1/2}$ (B) $\frac{\mu_0}{2\pi d}\left(I_1^2+I_2^2\right)^{1/2}$ (C) $\frac{\mu_0}{2\pi d}(I_1+I_2)$ (D) $\frac{\mu_0}{2\pi d}\left(I_1^2+I_2^2\right)$

3. Relative permittivity and permeability of a material are ε_r and μ_r, respectively. Which of the following values of these quantities are allowed for a diamagnetic material ?

(A) $\varepsilon_r = 1.5, \mu_r = 0.5$ (B) $\varepsilon_r = 0.5, \mu_r = 0.5$ (C) $\varepsilon_r = 1.5, \mu_r = 1.5$ (D) $\varepsilon_r = 0.5, \mu_r = 1.5$

4. A current-carrying, straight wire is kept along the axis of a square loop carrying a current. The straight wire

(A) will exert an inward force on the square loop

(B) will exert an outward force on the square loop

(C) will not exert any force on the square loop

(D) will exert a force on the square loop parallel to itself.

5. A circular loop is kept in that vertical plane which contains the north-south direction. It carries a current that is towards south at the topmost point. Let A be a point on axis of the circle to the east of it and B a point on this axis to the west of it. The magnetic field due to the loop

(A) is towards east at A and towards west at B

(B) is towards west at A and towards east at B

(C) is towards east at both A and B

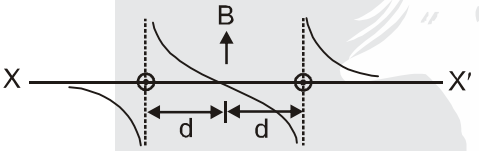
(D) is towards west at both A and B

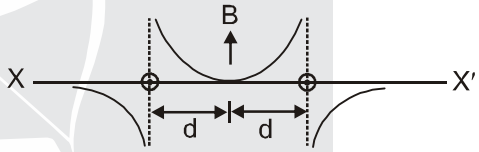
6. A toroid of mean radius ' a ' , cross section radius ' r ' and total number of turns N. It carries a current ' i '. The torque experienced by the toroid if a uniform magnetic field of strength B is applied :

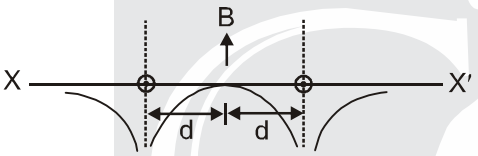
(A) is zero (B) is $B\,i\,N\,\pi\,r^2$

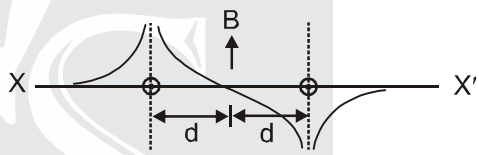
(C) is $B\,i\,N\,\pi\,a^2$ (D) depends on the direction of magnetic field.

7. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of magnetic field B along the line XX´ is given by

(A) 

(B) 

(C) 

(D) 

8. A current I flows in an infinitely long wire with cross–section in the form of a semicircular ring of radius R. The magnitude of the magnetic induction along its axis is :

(A) $\frac{\mu_0 I}{\pi^2 R}$ (B) $\frac{\mu_0 I}{2\pi^2 R}$ (C) $\frac{\mu_0 I}{2\pi R}$ (D) $\frac{\mu_0 I}{4\pi R}$

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9. An electric charge +q moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$, in an electromagnetic field given by :

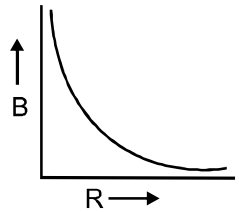
$\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} + 3\hat{k}$. The y - component of the force experienced by + q is :

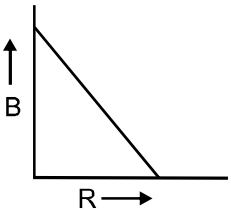
(A) 7 q (B) 5 q (C) 3 q (D) 2 q

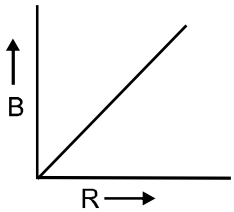
10. A thin circular disk of radius R is uniformly charged with density $\sigma > 0$ per unit area. The disk rotates about its axis with a uniform angular speed ω. The magnetic moment of the disk is :

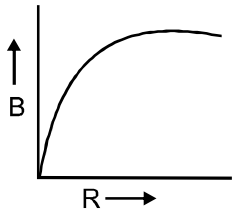
(A) $\pi R^4 \sigma\omega$ (B) $\frac{\pi R^4}{2}\sigma\omega$ (C) $\frac{\pi R^4}{4}\sigma\omega$ (D) $2\pi R^4 \sigma\omega$

11. A charge Q is uniformly distributed over the surface of non-condcting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω. As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. if we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure :

(A) 

(B) 

(C) 

(D) 

12. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is 2s. The magnet is cut perpendicular to its length into three equal parts and three parts are then placed on each other with their like poles together. The time period of this combination will be

(A) 2s (B) 2/3 s (C) $2\sqrt{3}$ s (D) $2/\sqrt{3}$ s

13. The coercivity of a small magnet where the ferromagnet gets demagnetized is 3×10^3 Am^{-1}. The current required to be passed in a solenoid of length 20 cm and number of turns 100, so that the magnet gets demagnetized when inside the solenoid, is :

(A) 30 mA (B) 60 mA (C) 3 A (D) 6 A

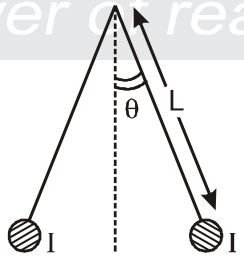
14. Two coaxial ideal and long solenoids of different radii carry current I in the same direction. Let $\vec{F_1}$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F_2}$ be the magnetic force on the outer solenoid due to the inner one. Then :

(A) $\vec{F_1} = \vec{F_2} = 0$ (B) $\vec{F_1}$ is radially inwards and $\vec{F_2}$ is radially outwards

(C) $\vec{F_1}$ is radially inwards and $\vec{F_2} = 0$ (D) $\vec{F_1}$ is radially outwards and $\vec{F_2} = 0$

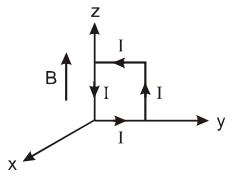
15. Two long current carrying thin wires, both with current I, are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle 'θ' with the vertical. If wires have mass λ per unit length then the value of I is : (g = gravitational acceleration)

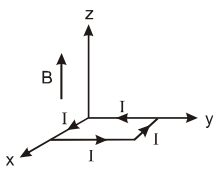
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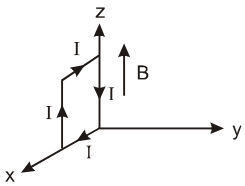


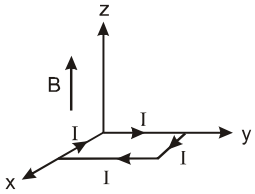
(A) $\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$ (B) $2\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$ (C) $2\sqrt{\frac{\pi gL}{\mu_0}\tan\theta}$ (D) $\sqrt{\frac{\pi\lambda gL}{\mu_0}\tan\theta}$

16. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below :

(a) 

(b) 

(c) 

(d) 

If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium ?

(A) (a) and (b), respectively (B) (a) and (c), respectively

(C) (b) and (d), respectively (D) (b) and (c), respectively

17. Two identical wires A and B, each of length l, carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side 'a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is :

(A) $\frac{\pi^2}{16\sqrt{2}}$ (B) $\frac{\pi^2}{16}$ (C) $\frac{\pi^2}{8\sqrt{2}}$ (D) $\frac{\pi^2}{8}$

18. A magnetic needle of magnetic moment 6.7×10^{-2} Am^2 and moment of inertia 7.5×10^{-6} kg m^2 is performing simple harmonic oscillations in a magnetic field of 0.01 T. Time taken for 10 complete oscillations is :

(A) 8.76 s (B) 6.65 s (C) 8.89 s (D) 6.98 s

19. The dipole moment of a circular loop carrying a current I, is m and the magnetic field at the centre of the loop is B_1. When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of loop is B_2. The ratio $\frac{B_1}{B_2}$ is :

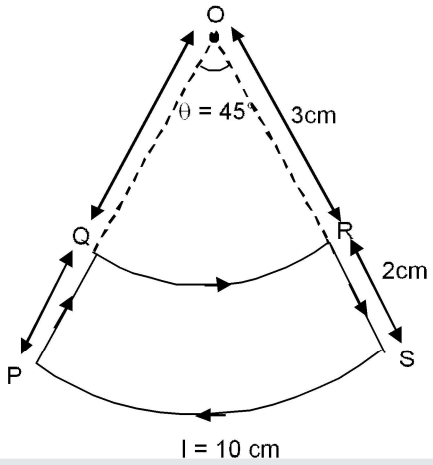
(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) 2 (D) $\sqrt{3}$

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20. An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e, r_p, r_α respectively in uniform magnetic field B. The relation between r_e, r_p, r_α is :

(A) $r_e < r_p < r_\alpha$ (B) $r_e < r_\alpha < r_p$ (C) $r_e > r_p = r_\alpha$ (D) $r_e < r_p = r_\alpha$

21. A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to :



(A) 1.5×10^{-5} T (B) 1.0×10^{-7} T (C) 1.0×10^{-5} T (D) 1.5×10^{-7} T

22. A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity of the bar magnet is :

(A) 285 A/m (B) 520 A/m (C) 1200 A/m (D) 2600 A/m

23. A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of magnetic filed of 0.5 T. If an electric field of 100 V/m makes it to move in a straight path, then the mass of the particle is (Given charge of electron = 1.6×10^{-19} C)

(A) 2.0×10^{-24} kg (B) 1.6×10^{-27} kg (C) 9.1×10^{-31} kg (D) 1.6×10^{-19} kg

24. An insulating thin rod of length ℓ has a linear charge density $\rho(x) = \rho_0 \frac{x}{\ell}$ on it. The rod is rotated about an axis passing through the origin (x = 0) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is :

(A) $\frac{\pi}{4} n\rho\ell^3$ (B) $\pi n\rho\ell^3$ (C) $\frac{\pi}{3} n\rho\ell^3$ (D) $n\rho\ell^3$

25. Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is

(A) $500\sqrt{3}$ (B) $250\sqrt{3}$ (C) $50\sqrt{3}$ (D) $5\sqrt{3}$

ANSWER KEY

1.	(A)	**2.**	(B)	**3.**	(A)	**4.**	(C)	**5.**	(C)
6.	(A)	**7.**	(A)	**8.**	(A)	**9.**	(A)	**10.**	(C)
11.	(A)	**12.**	(B)	**13.**	(D)	**14.**	(AD)	**15.**	(B)
16.	(C)	**17.**	(C)	**18.**	(B)	**19.**	(A)	**20.**	(D)
21.	(C)	**22.**	(D)	**23.**	(A)	**24.**	(A)	**25.**	(A)